Todd Takala

Data Scientist – Reliability Analytics – Fleet Analytics – Machine Learning Engineer – AI Engineer – Thought Leader
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Summary

Experienced data scientist with a strong background in reliability engineering, specializing in advanced analytics and machine learning to enhance asset availability and reduce costs. Proven track record in leading projects that delivered over $1.2B in value through AI innovations. Committed to leveraging expertise in data science to drive strategic initiatives and develop impactful data products.

Technical Skills

Programming Languages: Python, R, SQL, MATLAB, Bash, Powershell

Data Science, Data Engineering & Machine Learning: MLOps, Artificial Intelligence, AI Ethics, Snowflake Database Administrator, DBA, IoT, Condition Monitoring, ETL, Linux, AWS, Sagemaker, Step Functions, Inference, Redshift, PostreSQL, STAR Schema, Regular Expressions, Jupyter Notebooks

Frameworks: Autogluon, TensorFlow, PyTorch, Keras, ARIMA, SVM, Regression, Correlation, Classification, Clustering, Optimization, A/B Testing, Object-Oriented Programming (OOP), Random Forest, Tree-based Methods, GPU processing, CTE, Common Table Expression, Window Function

Libraries & Tools: NumPy, Pandas, Scikit-learn, Git, Docker, Tableau, Power BI with Advanced Data Visualization, ggplot2, matplotlib, seaborn, dplyr, tidyverse, caret, RJAGS, LightGBM, XGBoost cuDF, RAPIDS, CUDA

Experience

Lead Data Scientist, Service Capability — September 2022 – Present
Caterpillar Inc. — *Remote*

• An artificial intelligence-based system was developed to estimate repair times for Caterpillar equipment by utilizing machine learning algorithms applied to warranty and service data. This system contributed to a reduction in warranty costs amounting to $80M in 2024. The project received the CEO Award for its significant impact, employing advanced tools and methodologies such as Python, R, SQL, Snowflake, Autogluon, neural networks, random forest, TensorFlow, PyTorch, HuggingFace, natural language processing techniques, A/B testing, and statistical inference.

• Designed and deployed robust ETL/MLOps pipelines and advanced ML/AI workflows using Snowflake as a database administrator & developer, AWS S3 for data storage, and AWS Step Functions to orchestrate automated model retraining and deployment. Utilized AWS SageMaker for model development, ECR for containerization, and Secrets Manager to ensure secure credential management across all environments.

• Skilled in explaining complex conclusions to non-technical partners. Resolved a major challenge of unclear project specifications for a remote team of six by creating a structured approach for gathering and integrating stakeholder input. This led to clear agile user stories, reducing misunderstandings and improving team cohesion and project performance.

• Acted as the technical steward of DevOps, Github, Snowflake, Docker, and AWS architecture, setting code standards, best practices, and conducting research and development. Monitored cloud usage, scaling, tuning, and optimized cloud cost while maintaining performance.

Reliability Analytics Manager, Amazon Robotics — February 2022 – September 2022
Amazon.com — *Remote*

• Secured two-time champion status in Amazon Machine Learning University contests by developing highly accurate regression and recommendation systems. This success was a result of an in-depth understanding of advanced ML algorithms, feature engineering techniques, and model hyperparameter tuning and optimization best practices.

• Initiated an effort to establish a new design-for-reliability (DfR) benchmark by analyzing IoT data from 400,000 robots using an advanced technology stack. Parsed firmware log time series data, queried extensive tables with Redshift and PostgreSQL, and extracted data from Glacier and S3, accumulating over 2 petabytes of information. Authored a research paper that highlighted deficiencies in the current design life metric and conducted a Weibull statistical analysis, leading senior leadership to agree on its adoption. Effectively communicated complex analytical concepts to non-technical stakeholders.

Data Engineering Manager – Reliability Engineering Manager January 2012 – February 2022
Empire-Cat *Mesa, AZ*

• Developed a regression model to ascertain the principal factors impacting the various stages of equipment lifecycle and safety, resulting in cost savings of $62M in the first 90 days. Integrated data from work orders, failure analyses, parts information, fluid chemistry, and IoT telemetry to create a comprehensive machine learning training dataset for ongoing optimization. (R, SQL, NLP, Leadership, Diesel Engines)

• Resolved critical design issues in mining equipment, cutting client operating costs by $1.2B over 6 years. Achieved and maintained 90% physical availability for a legacy fleet of 250 haul trucks, setting an exceptional benchmark in the mining industry by addressing failures in engines, transmissions, drivetrain, and hydraulics components.

• Led cross-functional collaboration and client-facing problem-solving, translating complex technical and mathematical concepts into clear, actionable user stories. Enhanced communication and stakeholder engagement across all organizational levels, resulting in improved project outcomes and client satisfaction.

• Applied time series forecasting modeling methods, such as ARIMA and SVM, to detect trends and patterns within the data. Deployed an MLOps pipeline to analyze trends in condition monitoring, IoT, and telematics data of machine components, forecasting demand for the re-manufacturing supply chain. Implemented using Python, R, SQL, and Power BI.

Technical Consultant, Reliability Engineering January 2008 – January 2012
Road Machinery, LLC *Phoenix, AZ*

• Provided expert-level root cause failure analysis (RCFA) for intricate failures in high-value internal combustion engines and powertrain components. Synthesized findings into detailed summaries that translated complex technical issues into actionable strategies, leading to improved maintenance practices and product enhancements for engineering teams and external clients

• Developed an advanced total cost of ownership (TCO) projection for a fleet of 18 vehicles by integrating and synthesizing diverse datasets using MATLAB and SQL. The analysis quantified the long-term financial advantages, increased speed on grade, reduced maintenance costs, and enhanced operational efficiencies, which served as the pivotal financial justification for securing a $72M contract.

• Generated $2.5M in consulting revenue over 3.5 years by delivering expert insights that optimized client mining fleets. Recommendations on asset life cycle management and engineering cost and finance directly led to augmented operational efficiency and a stronger financial position for clients, securing high-value, long-term contracts.

Research Development Design Engineer June 2006 – December 2007
Komatsu America Corp. *Chattanooga, TN*

• Upgraded the longevity of mechanical systems by integrating findings from the FRACAS process into a comprehensive redesign workflow. Utilized FEA (Finite Element Analysis) to simulate stress points and identify failure modes, and then leveraged CAD (Computer-Aided Design) tools to implement step-by-step design modifications that effectively resolved system weaknesses, resulting in enhanced system durability.

• Directed the end-to-end product reliability and quality enhancement lifecycle. This involved creating and testing multiple prototypes for new designs, assessing performance data from multiple sources, and executing targeted corrective actions and design enhancements using FEA and CAD tools, ensuring a robust and reliable final product.

EDUCATION

Arizona State University June 2003 – May 2006
B.S. in Engineering, Mechanical Engineering *Tempe, AZ*

U.S. Department of Labor & IUOE Local 150 Apprenticeship and Skill Improvement Program April 1998 – April 2001
Vocational Certificate, Operating Engineer *Washington, DC*

CERTIFICATIONS

SQL Associate Certificate: Datacamp.com January 2024

Datacamp.com December 2023

AI Fundamentals certificate: Datacamp.com September 2023

Lean 6 Sigma Green Belt: Empire-Cat February 2013

Certified Technical Communicator: Komatsu North America October 2011

EIT, Mechanical Engineering: Arizona State Board of Technical Registration – Registration No. 10670 January 2009